

Special Seminar on Multi-Modal AI

Jiayuan Mao

Department of CIS, GRASP

<https://cis7000-fall26.thelabone.org/>



Penn
Engineering
UNIVERSITY of PENNSYLVANIA

FA26

Lecture 10: Video Understanding

Special Seminar on Multi-Modal AI

Jiayuan Mao

Department of CIS, GRASP

<https://cis7000-fall26.thelabone.org/>



Goal of This Lecture

Use Machine Learning to Understand Videos

Goal of This Lecture

Use machine learning to understand videos



Formulating Video Understanding

Task	Input	Output
Action recognition	A short clip	An action class
Temporal localization	A video and an action	Start and end times
Video–text retrieval	A text query and videos	A ranked list of videos
Video question answering	Video, question, optional subtitles	An answer

Overview

Temporal representation: motion, ordered relations, and attention
Two-Stream, I3D, TRN, TSM, SlowFast, TimeSformer

Learning without action labels: choose a prediction target
VideoMAE and V-JEPA

Connecting videos to language: InternVideo2

Testing temporal perception and long-video reasoning
LongVideoBench, T*

Video Classification: Inputs and Labels



(g) riding a bike



(h) riding unicycle

Kinetics-400: a short human-action clip with one of 400 class labels

These examples have labels “riding a bike” and “riding unicycle”

Prediction: class scores for the clip Metric: top-1 / top-5 accuracy

Similar Objects, Different Actions



(m) dribbling basketball



(n) dunking basketball

The labels distinguish “dribbling basketball” from “dunking basketball”

Useful temporal evidence depends on the specific class distinction

Something-Something: Object Operations



Input: a short clip of an object interaction

Label: “Picking [something] up”
The object here is a shoe

Predict the action template, rather than the object name

V1 and V2 share 174 action categories but have different data splits

A Baseline: Recognize Frames Independently

Apply one image encoder to every sampled frame

$$z = \frac{1}{T} \sum_{t=1}^T f(I_t)$$

I : frame f : image encoder T : sampled frames z : video feature

Average the frame features, then predict an action

Any permutation of the same frames gives the same average

The model cannot distinguish order using this representation alone

Two-Stream ConvNet: Action Classification



Input: a video clip

Output: an action class

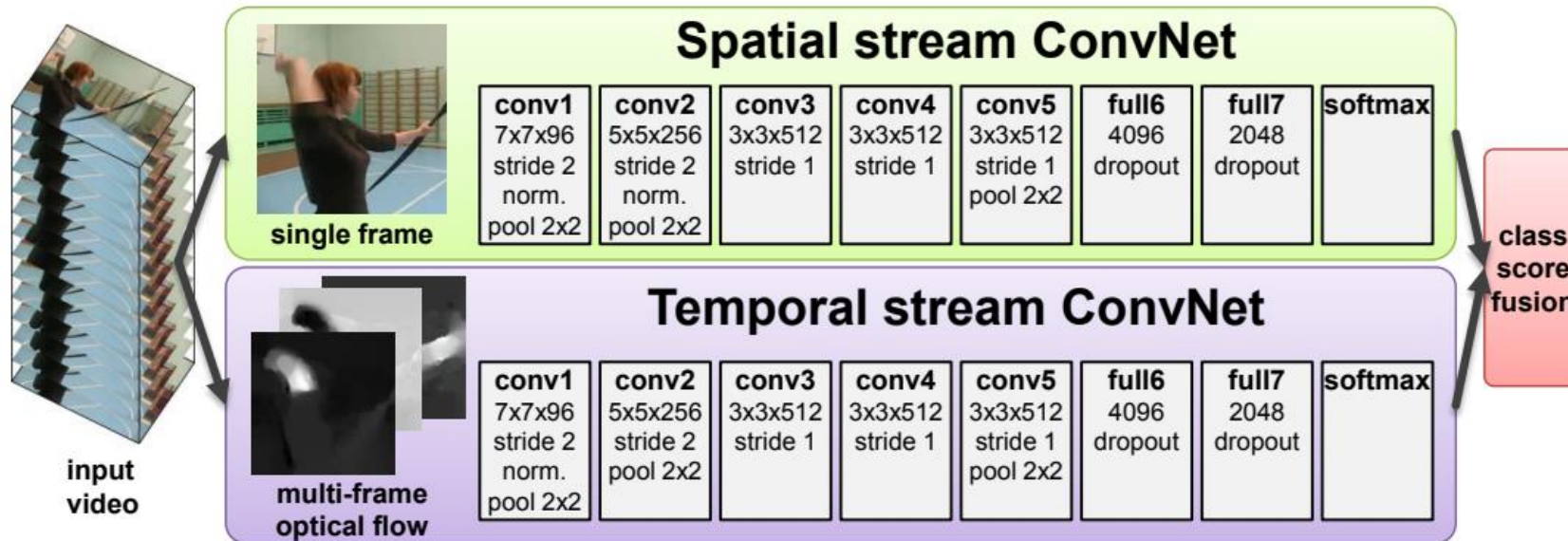
UCF101 has 101 classes

HMDB51 has 51 classes

Training uses clip labels

Evaluation uses accuracy

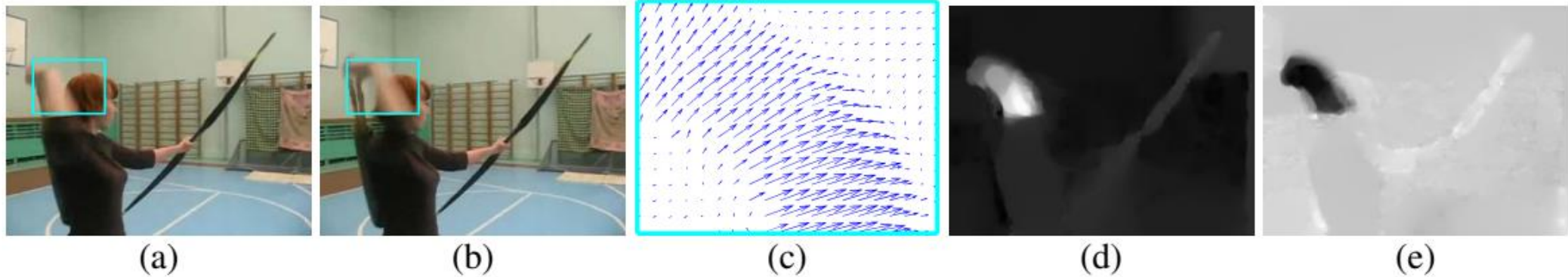
Two-Stream: Appearance and Motion



Key idea: use an RGB stream and a stream of precomputed optical flow

Learn action classifiers for both inputs, then combine their class scores

Optical Flow: What the Second Stream Sees

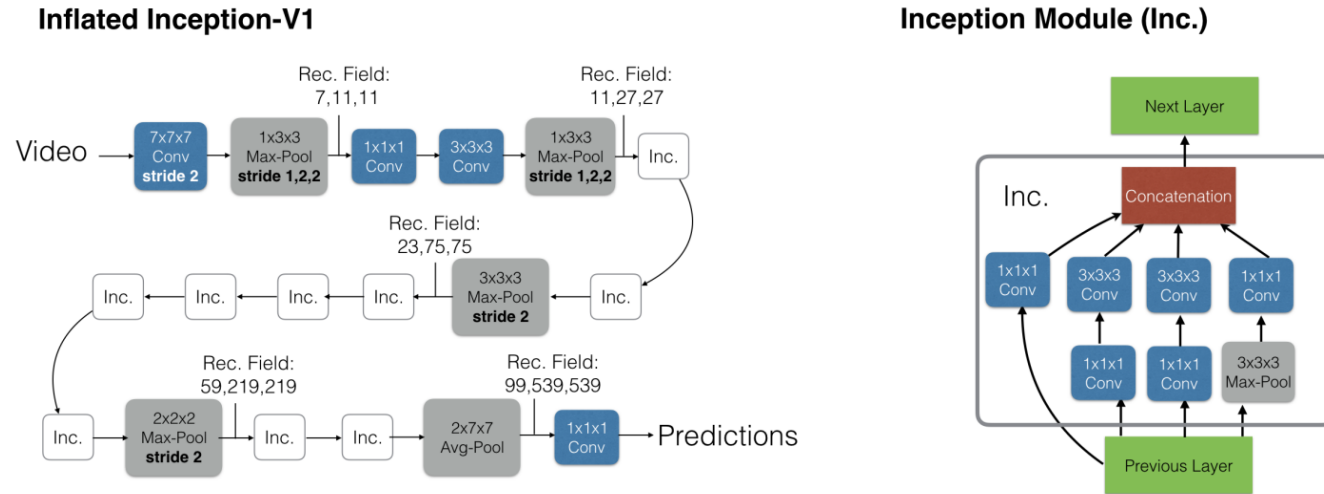


Flow estimates the horizontal and vertical displacement at each pixel

Ten consecutive flow fields become twenty input channels

Limit: camera motion and flow estimation errors also enter this stream

I3D: Learning a Spatiotemporal Classifier



Input: RGB or optical-flow clips Output: action-class scores

Key idea: extend an image CNN with a temporal dimension in its filters

Train on Kinetics labels · Evaluate action accuracy on Kinetics, UCF101, HMDB51

Temporal Interaction Before Pooling

A temporal filter combines features from neighboring times

$$y_t = w_{-1}x_{t-1} + w_0x_t + w_{+1}x_{t+1}$$

x: frame feature w: learned temporal weight t: time index

Different weights for past and future can encode direction

3D convolutions learn spatial and temporal filters together

I3D: Initializing Time from an Image Filter

$$W_{3D}[\tau] = \frac{W_{2D}}{K}$$

Copy the 2D filter into K temporal positions and divide each copy by K

TRN: Action Recognition from Sparse Frames

Input: a short RGB video of an object interaction

Pretending to put something into something



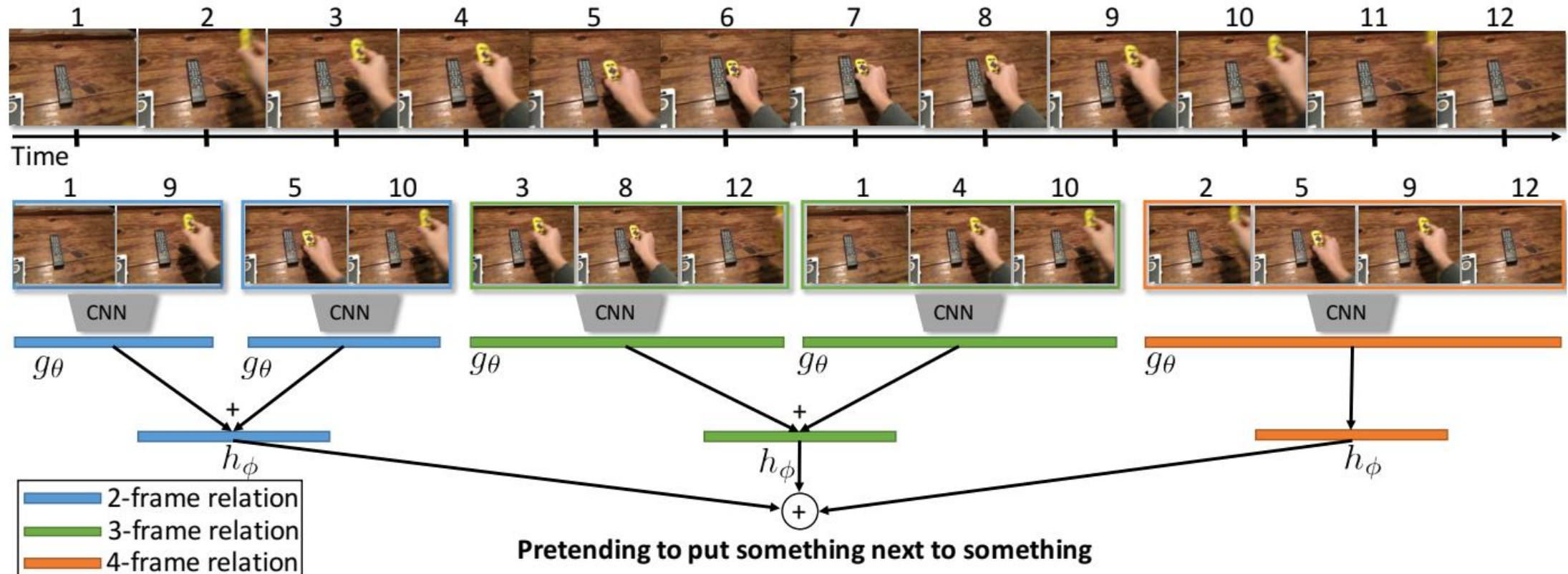
1: Pretending to put sth into sth (0.710). 2: Failing to put sth into sth because sth does not fit (0.265)

Output: one of 174 action labels

Example: “Pretending to put something into something”

Something-Something V1 / V2 · Clip labels for training · Top-1 accuracy

TRN: Combining Ordered Observations



Sample frames across the clip and reuse their CNN features

Key idea: learn relations among ordered pairs, triples, and larger tuples

TRN: A Relation Function

$$T_2(V) = h_{\varphi}(\sum_{i < j} g_{\theta}(f_i, f_j))$$

f_i : frame i feature g_{θ}, h_{φ} : learned MLPs

Concatenate each earlier–later pair before applying the MLP

$$MT_8(V) = \sum_{d=2}^8 T_d(V)$$

Each relation size has its own learned parameters

Changing the order can change the class scores

TRN: Evidence for Temporal Relations

Dataset	Average pooling	Relation pooling
UCF101	82.69%	83.83%
Something-Something V1	19.53%	34.44%
Jester	85.41%	95.31%

Top-1 accuracy with the same BN-Inception backbone and 8 RGB frames

The gain depends on the task's need for temporal evidence

TRN: Which Frames Support the Label?



Later observations show that the apparent opening never completes

Sampling limit: sparse frames can miss a brief opening followed by closing

TSM: Problem Setting

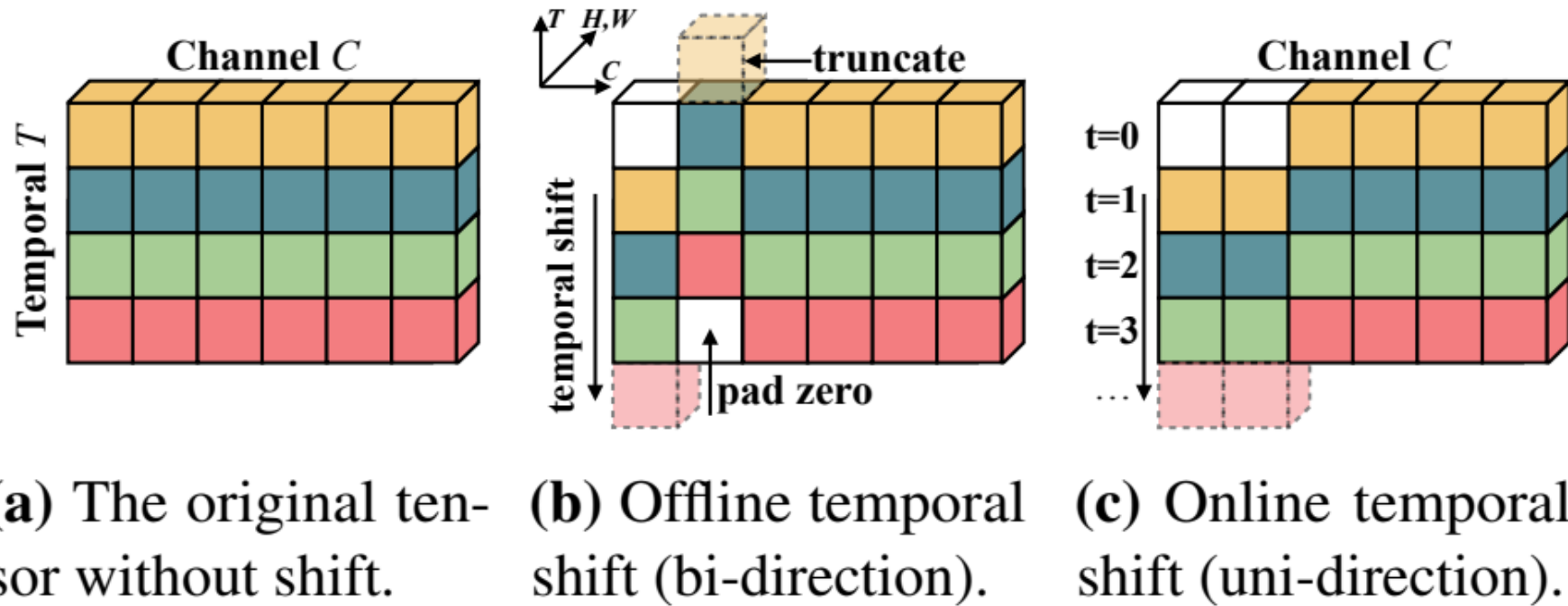
Input: a sampled sequence of RGB frames from one clip

Output: a probability for each action category

Research question: can a 2D CNN exchange temporal evidence cheaply?

Evaluation: accuracy together with computation and latency

TSM: Exchange Features Across Time



Key idea: shift some feature channels before a learned 2D convolution

The convolution can now mix past, present, and future information

TSM: A Worked Channel Example

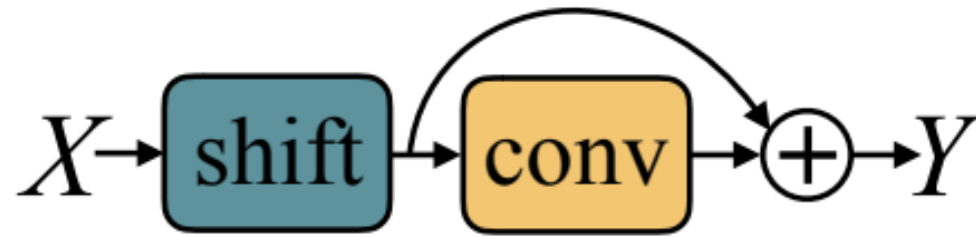
At one time step, divide eight channels into three groups

Channel group	Before shifting	After shifting
1 channel	Current frame	Previous frame
6 channels	Current frame	Current frame
1 channel	Current frame	Next frame

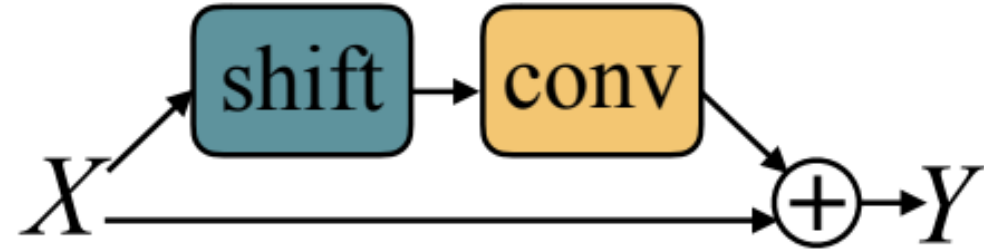
The next convolution mixes these groups using learned weights

Stacking shifted blocks expands the temporal receptive field

TSM: Preserve the Current Frame



(a) In-place TSM.



(b) Residual TSM.

Shift inside the residual branch

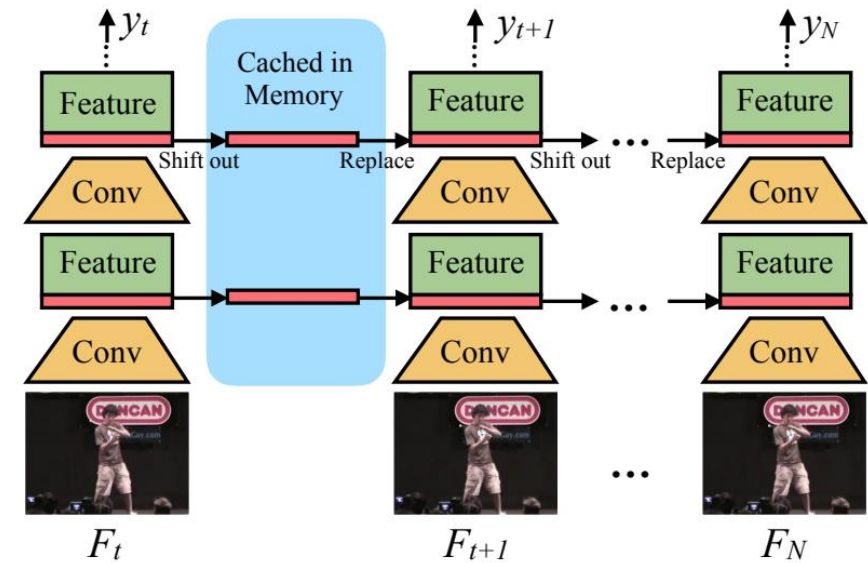
The identity connection keeps the original current-frame features available

Partial shifting also reduces memory movement

TSM: Offline and Online Inference

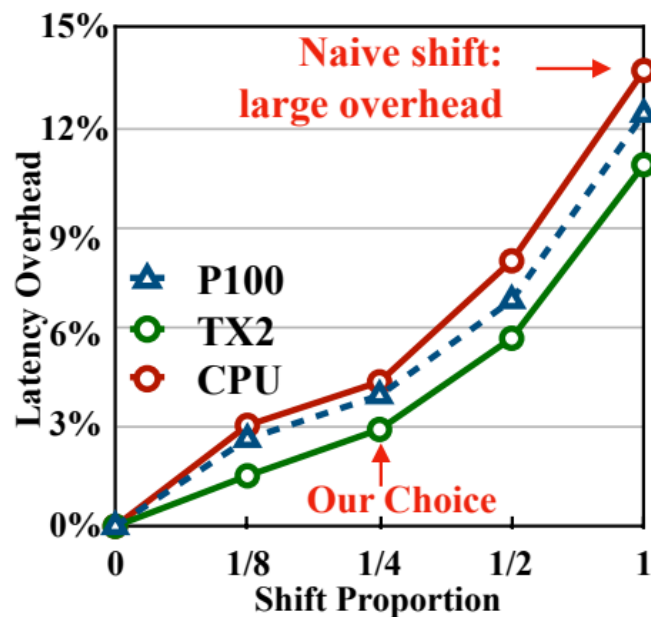
Offline recognition can use frames on both sides of the current time

Online recognition caches one eighth of the previous features at each block

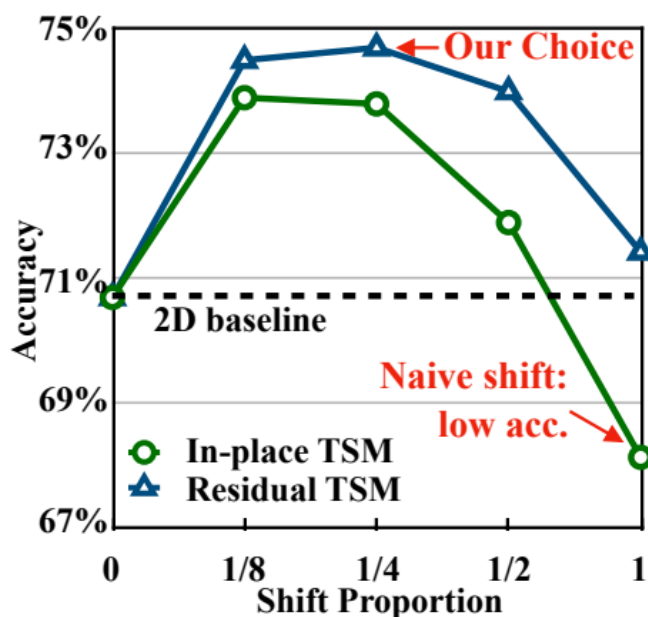


A live system cannot wait for future frames that have not arrived

TSM: Zero FLOPs Still Has a Cost



(a) Overhead vs. proportion.



(b) Residual vs. in-place.

Moving activations consumes memory bandwidth

Shifting too many channels can weaken spatial features

SlowFast: Problem Setting

Input: RGB frames sampled at two rates from the same time interval

Output: action-class scores for that interval

Training: labeled action videos

Kinetics uses one label per clip and cross-entropy

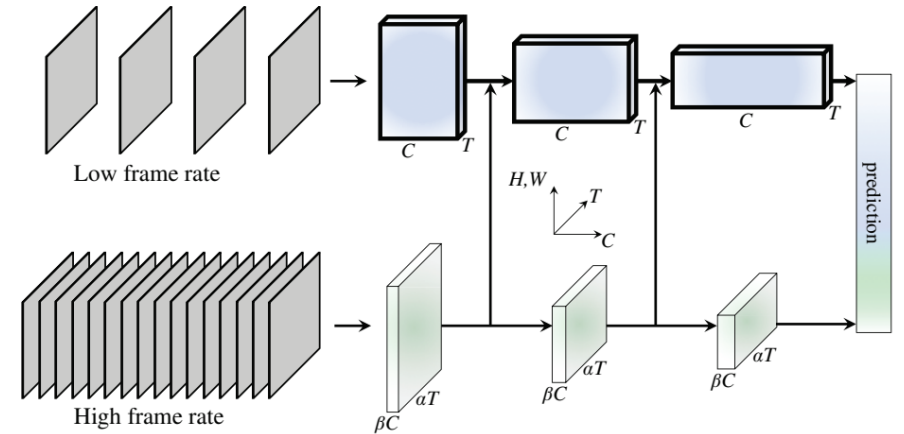
The same backbone can also support person-level action detection

AVA adds person boxes and allows multiple action labels

SlowFast: Allocate Capacity by Time Scale

Key idea: use a slow, wide pathway for appearance and a fast, narrow pathway for motion

Both pathways receive RGB frames from the same time interval



Learned lateral connections let the pathways exchange information

SlowFast: One Concrete Configuration

Start with the same 64-frame raw clip

	Slow pathway	Fast pathway
Sampled frames	4	32
Sampling stride	16	2
Relative channels	1	1/8

The Fast pathway samples eight times as densely

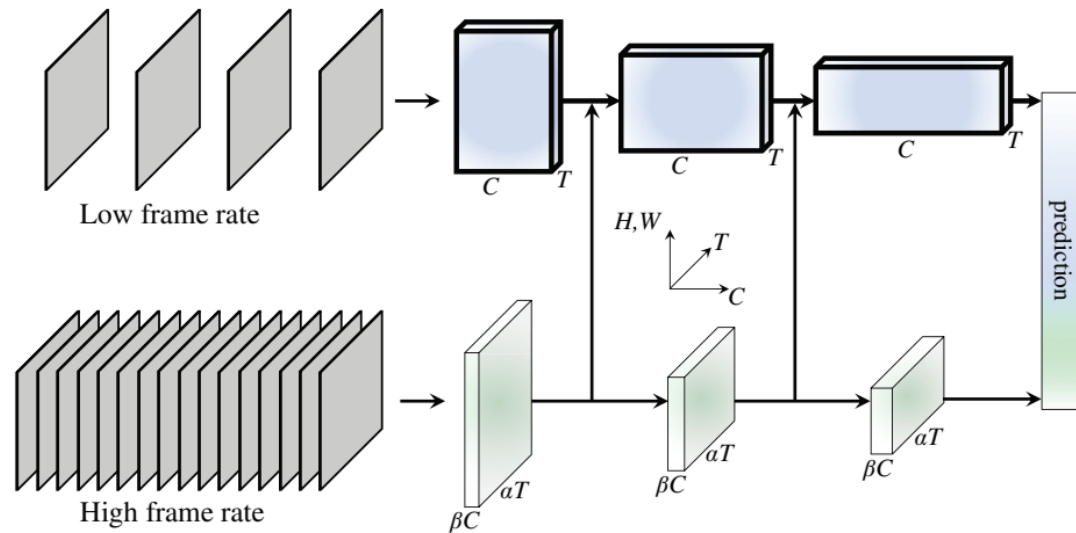
Why Can the Fast Pathway Be Lightweight?

Convolution cost often scales with both input and output channel counts

$$\text{relative cost} \approx \alpha \beta^2 = 8 \times \left(\frac{1}{8}\right)^2 = \frac{1}{8}$$

α : frame-rate ratio β : channel-width ratio

SlowFast: Fusion Across the Hierarchy



Align the Fast features to the Slow time grid

The default lateral transform uses a temporal convolution with stride eight

Concatenate aligned features at intermediate stages

SlowFast: Learning and Evaluation

Learn both pathways and their fusion jointly from video labels

Kinetics action recognition predicts one class for the clip

The paper trains these models from scratch

SlowFast: Complementary Pathways

Kinetics-400, 4×16 ResNet-50 configuration

Configuration	Top-1 accuracy	GFLOPs / view
Slow only	72.6%	27.3
Fast only	51.7%	6.4
Both, final fusion only	73.5%	34.2
Both, temporal-convolution fusion	75.6%	36.1

The Fast pathway helps more when fusion occurs throughout the network

Does the Model Need Time?

A model recognizes “playing tennis” from the court and racket

For evaluation, we can design two clips with the same objects but different correct action labels

Overview

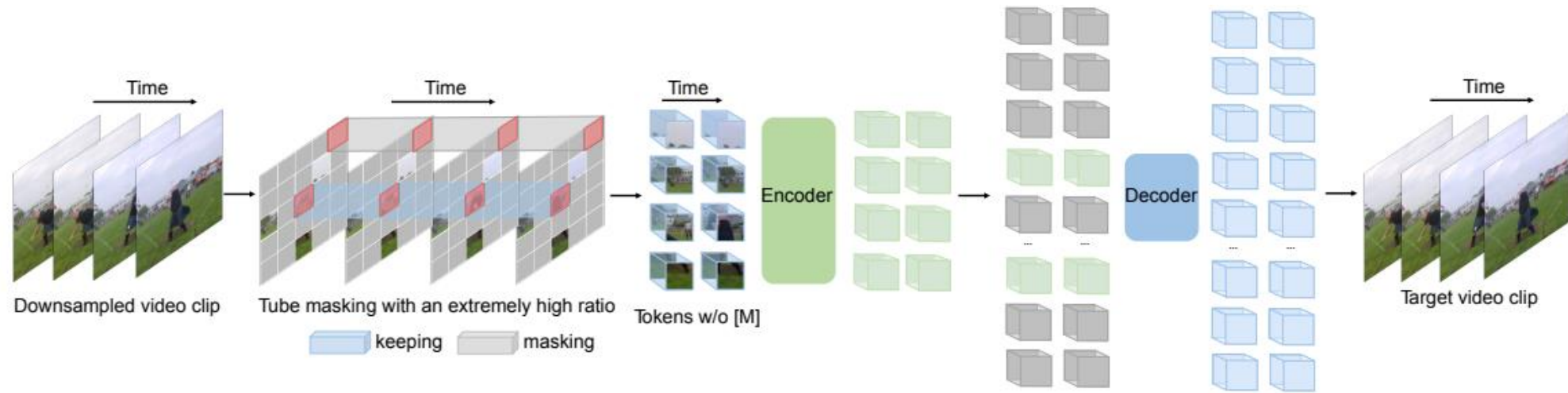
Temporal representation: motion, ordered relations, and attention
Two-Stream, I3D, TRN, TSM, SlowFast, TimeSformer

Learning without action labels: choose a prediction target
VideoMAE and V-JEPA

Connecting videos to language: InternVideo2

Testing temporal perception and long-video reasoning
LongVideoBench, T*

VideoMAE: Learning Before Action Labels

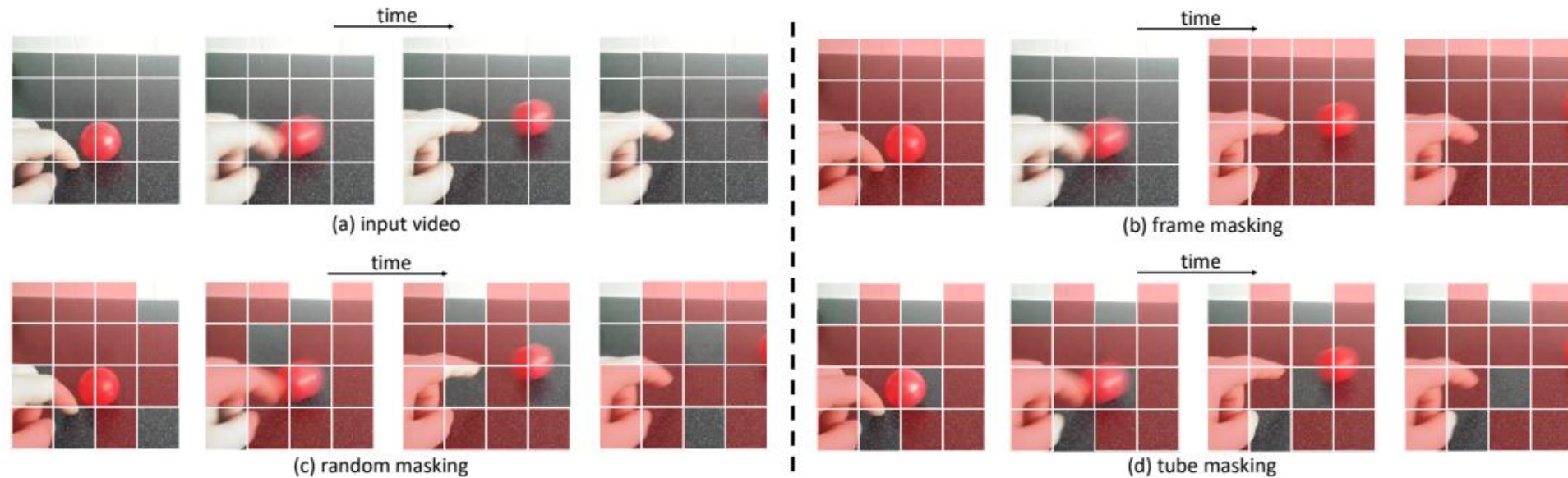


Pretraining input: a video with most tokens removed

Target: the missing video pixels

Downstream task: attach a classifier and fine-tune using action labels

VideoMAE: Tube Masking?



Use the same spatial mask across time and hide 90–95% of tokens

Key idea: reduce shortcuts from copying nearby visible content

VideoMAE: Objective and Evaluation

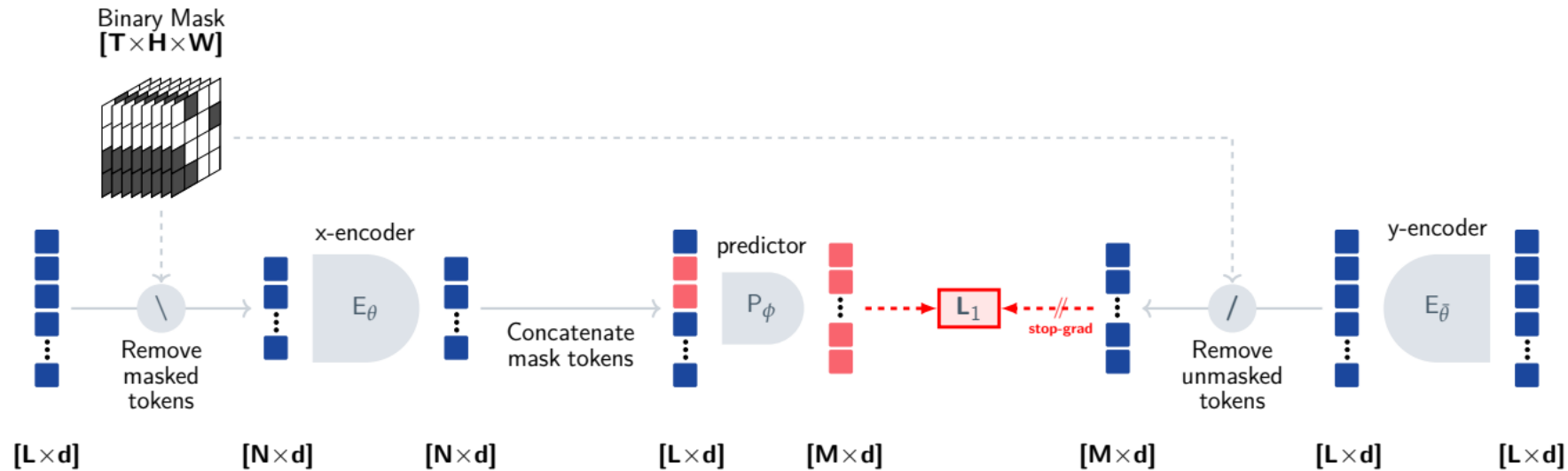
$$L = \frac{1}{|M|} \sum_{p \in M} \|\hat{x}_p - x_p\|^2$$

M: masked tubelets x: target pixels $\hat{x}$: decoder prediction

Train the encoder and decoder by reducing masked-pixel reconstruction error

Limit: good reconstruction can preserve texture without solving the downstream semantic task

V-JEPA: Predicting Missing Features



Pretraining input: visible parts of a video

Target: encoder features at the missing positions

V-JEPA: What Is Learned and What Is Fixed?

$$L = \frac{1}{|M|} \sum_{p \in M} \|\hat{z}_p - \text{sg}(\bar{z}_p)\|_1$$

$\hat{z}$: predicted feature $\bar{z}$: target-encoder feature
sg: stop gradient M: masked positions

Update the context encoder and predictor with the loss
Update the target encoder with a moving average
Otherwise it will easily degenerate

V-JEPA: Testing the Frozen Representation

Freeze the pretrained video encoder

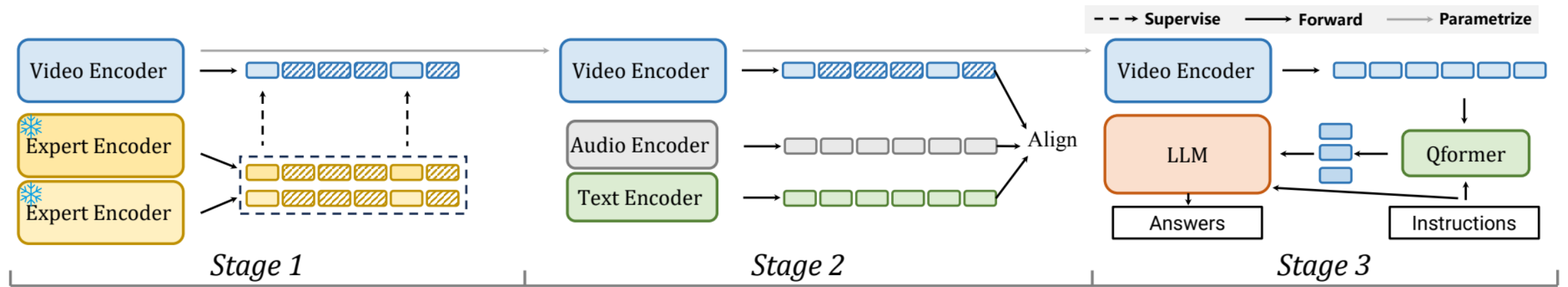
Train an attentive pooling head using downstream action labels

Report top-1 accuracy separately from full fine-tuning

Choosing the Prediction Target

Pretraining method	Prediction target	Where the target comes from
VideoMAE	Missing pixels	The observed training video
V-JEPA	Missing token features	A moving-average encoder
InternVideo2, stage 1	Expert features at visible tokens	Fixed pretrained teacher models

InternVideo2: Progressive Training



Key idea: progressively learn video features, multimodal correspondence, then dialogue

The video encoder grows to 6B parameters, with distinct training objectives at each stage

InternVideo2: The Video Encoder

Sample eight frames and divide each frame into 14×14 -pixel spatial patches

Represent the patches as tokens with spatial and temporal positions

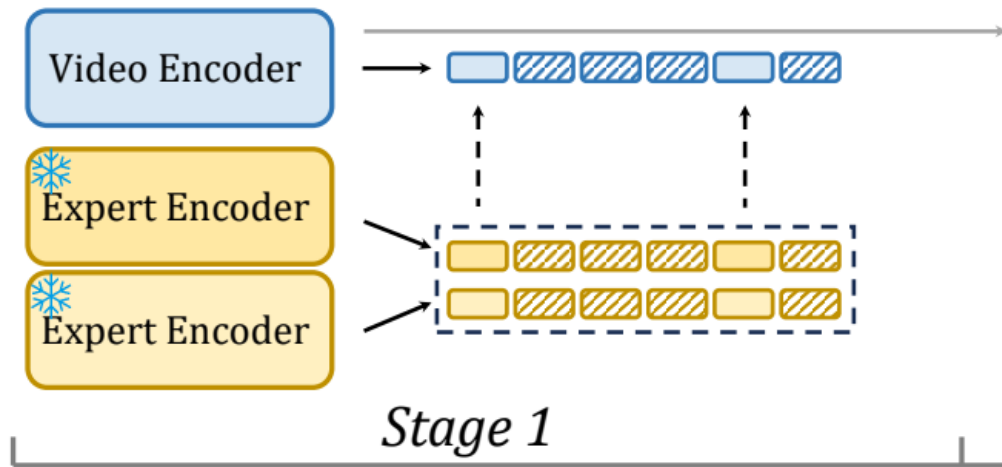
A Vision Transformer learns interactions among the tokens
Attention pooling produces a compact representation

Stage 1: Learn from Expert Video Features

Teachers see the full video

The student sees only 20% of the tokens

Match teacher features at the visible token positions



Key distinction: the targets are expert features at unmasked positions

Stage 1: Distilling Expert Features

$$L = \frac{1}{|U|} \sum_{p \in U} \sum_k \lambda_k \|P_k(s_p) - e_{k,p}\|^2$$

U: visible tokens s: student feature e: expert feature

P: learned projection λ : weight for each expert

Compare student features with fixed semantic and motion teachers

Projection layers learn to put the features into comparable spaces

After pretraining, discard the projection layers and keep the encoder

Stage 2: Align Video and Language

$$\ell_i = -\log \frac{\exp(s(v_i, t_i)/\tau)}{\sum_j \exp(s(v_i, t_j)/\tau)}$$

v: video embedding t: text embedding s: cosine similarity

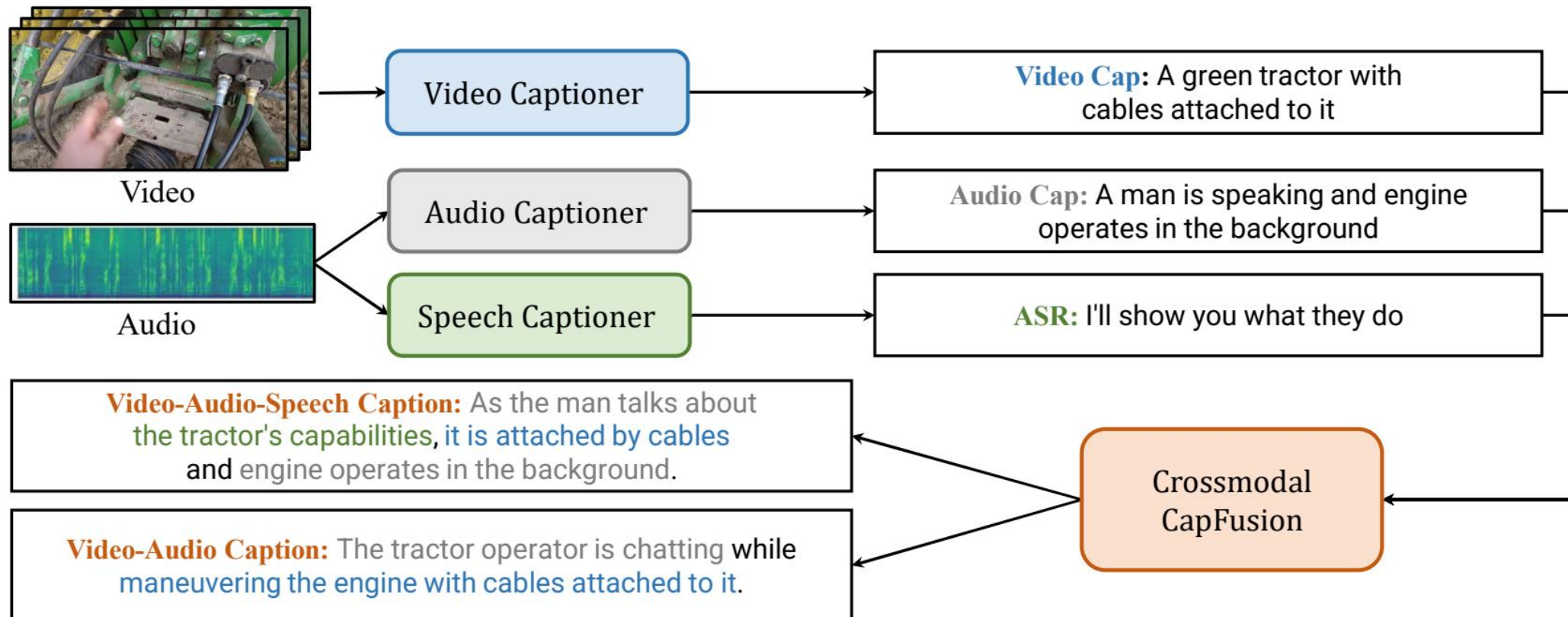
i: paired example j: candidate text τ : temperature

Raise similarity for paired video–text examples relative to other pairs in the batch

Also train pair matching and masked language reconstruction

Text connects image, video, and audio representations

InternVideo2: Constructing Multimodal Data



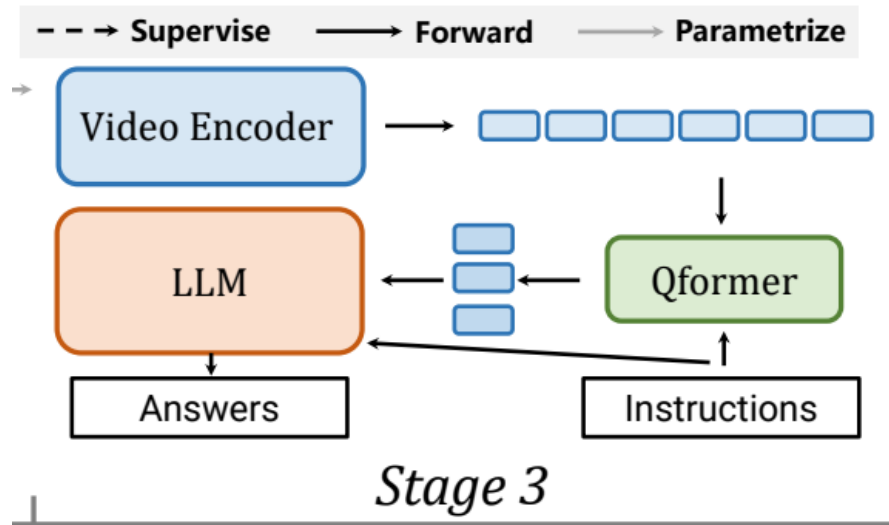
VidCap combines visual captions, audio captions, and speech transcripts

InternVideo2: What Data Enters Each Stage?

Stage	Training material	Reported scale
1: feature learning	K-Mash video collection	2M entries
2: multimodal alignment	Image–text pairs	300M
2: multimodal alignment	WebVid + InternVid video–text	About 50M
2: multimodal alignment	InternVid2 video–audio–speech–text	50M used
3: instruction tuning	Image/video conversations and QA	2.1M

The caption must refer to the sampled clip, not merely the source video

Stage 3: Connecting Features to Dialogue



A Q-Former converts video features into inputs for a language model

Train on video-centered instructions and answer text

$$L = - \sum_j \log p(a_j \mid a_{<j}, V, q)$$

a: answer tokens V: video q: question

Evaluating a Foundation Model

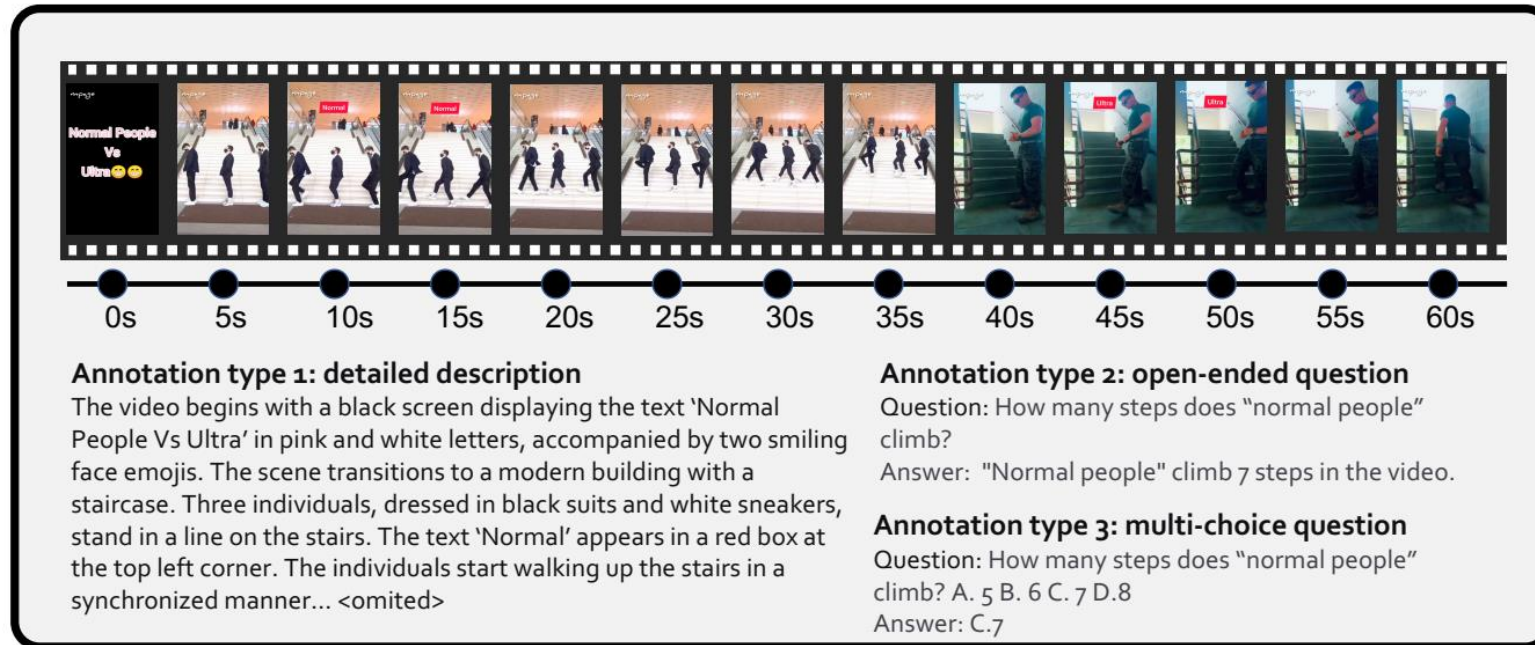
Setting	What changes on the target task?	What it tests
Linear / attentive probing	Train a small head, freeze the backbone	Usefulness of fixed features
End-to-end fine-tuning	Update the encoder and task head	Adaptability with target labels
Zero-shot retrieval	No target-benchmark training	Transfer of crossmodal alignment
Video dialogue / QA	Evaluate the connected language model	Answer quality under a prompt

A score belongs to a checkpoint, adaptation setting, and input protocol

InternVideo2: What Scaling Cannot Guarantee

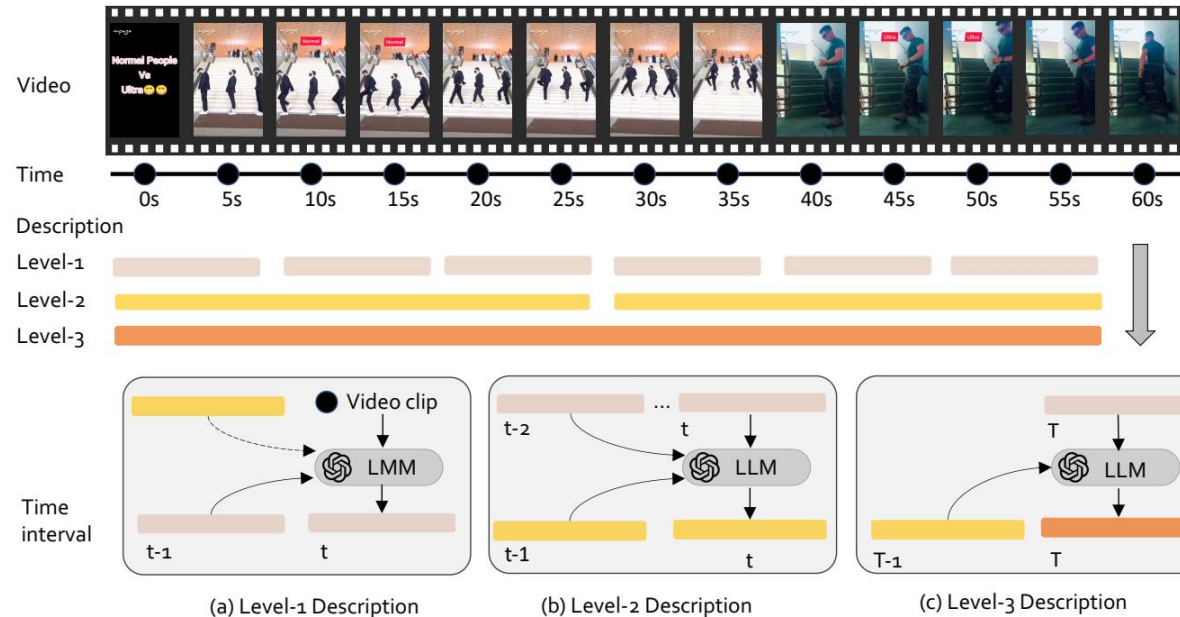
Fixed sampling can miss brief events before the encoder sees them

LLaVA-Video: Instruction Following



Input: video and instruction Output: a caption or an answer

LLaVA-Video: Building Training Data



Key idea: describe short intervals with history, then summarize across intervals

Generate question–answer pairs from the descriptions and filter them

LLaVA-Video: Training and Evaluation

LLaVA-Video-178K: 178K videos and about 1.3M instruction examples
Captions, open-ended answers, and multiple-choice answers

A visual encoder and projector supply video tokens to an LLM
Instruction tuning learns to predict the target answer tokens

Evaluate on captioning and QA benchmarks with their own scoring rules
Synthetic descriptions can miss events or introduce unsupported details

Language Supervision Has Different Outputs

Approach	Training relationship	At inference
Frozen in Time	A video matches a caption	Rank a candidate gallery
InternVideo2	Feature targets, modality pairs, then QA	Choose the task-specific checkpoint
LLaVA-Video	Input: A video and instruction Output: captions or answers	Generate an answer or caption

A detailed answer requires evidence that survived sampling and compression

The Cost of Keeping Every Frame

One hour at 30 frames per second contains 108,000 frames

Even one frame per second leaves 3,600 images

$$3,600 \times 256 = 921,600 \text{ visual tokens}$$

Illustration: 256 tokens per image, before subtitles or the question

Sampling and compression trade compute for potentially missing evidence

LongVideoBench: Problem Setting

Input: a long video, time-aligned subtitles, and a referring question

The evaluation samples frames under a specified input budget

Output: one answer choice

The model must identify the referred context and inspect its details

Evaluation: multiple-choice accuracy, including duration subsets

LongVideoBench: Referring Reasoning



Question:

At the beginning of the video, a woman with a headband tied to her head, wearing a red top, carrying a black backpack, when the woman comes down from a hill with tall rocks, what changes occur to her backpack?

Options:

- A. There is a dark red jacket hanging on her black backpack
- B. Nothing changed
- C. There is a white jacket hanging on her black backpack
- D. There is a dark blue jacket hanging on her black backpack**

Key idea: identify the referred context, then answer a detailed question about it

Perception questions refer to one moment

Relation questions connect multiple moments

LongVideoBench: Building the Benchmark

Collect videos and time-aligned English subtitles across ten content categories

Annotators watch each video, write a referring question, and mark relevant moments

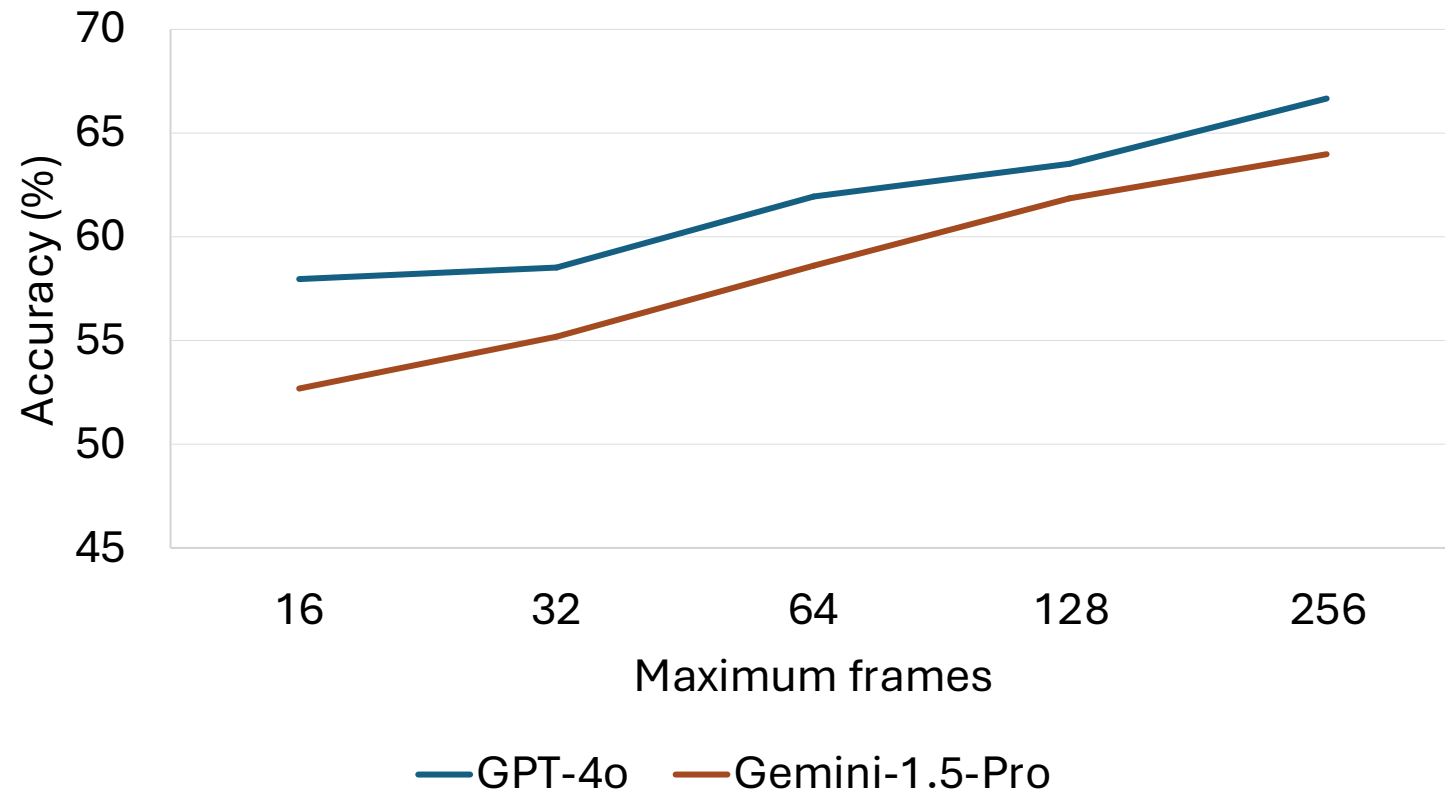
An examiner checks the annotation and a reviser fixes identified problems

LongVideoBench: Evaluation Protocol

Property	Protocol
Data	3,763 videos and 6,678 human-annotated questions
Duration groups	8–15 s, 15–60 s, 3–10 min, 15–60 min
Question types	17 categories across perception and relation
Validation / test	1,337 / 5,341 multiple-choice questions
Input	Sampled frames interleaved with aligned subtitle chunks
Metric	Multiple-choice accuracy

This is an evaluation benchmark only

More Frames Can Help or Hurt



Validation accuracy (%) vs. maximum frames, capped at 1 fps

Combining Frames and Subtitles

GPT-4o on the LongVideoBench validation set

Frames	Subtitles	Accuracy
No	Yes	44.6%
Yes	No	60.6%
Yes	Yes	66.7%

Combining modalities improves this model’s answers in this setting

Subtitles can identify when to look, while frames show visual details

Search for Evidence

Question: which few frames would be sufficient to answer?

Uniform sampling spends the same budget on relevant and irrelevant intervals

T* adapts the search to the question before asking the final answer model

T*: Evidence Search Before Video QA

Input: a long video and a question

Intermediate output: frames relevant to the question

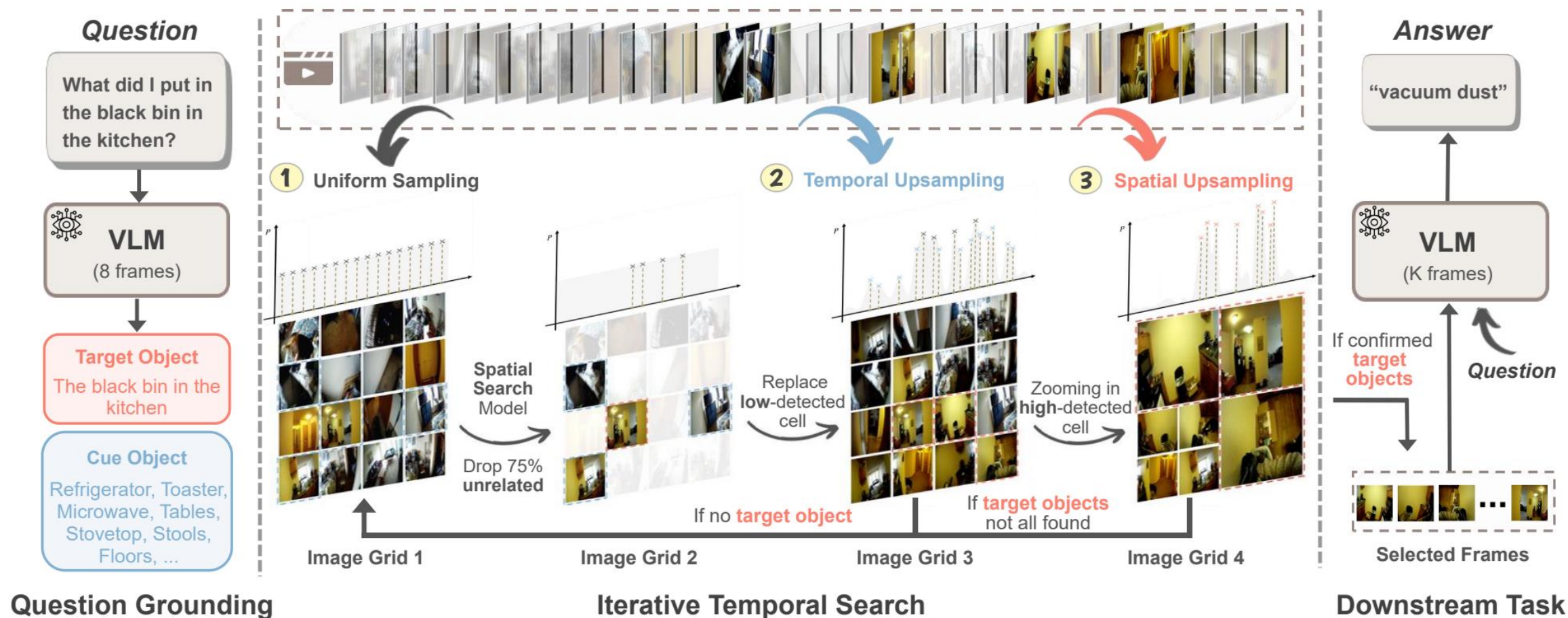
A downstream video-language model answers from the selected evidence

The search and answer stages have different budgets

Use existing multimodal models at inference time

Evaluate both evidence retrieval and downstream answer accuracy

T*: Recast Temporal Search as Spatial Search



Key idea: arrange frames in a grid, then search and refine promising regions

T*: What Changes During Search?

Ground the question into targets and contextual cues

Temporal: Score frames and increase sampling near promising frames

Spatial: Use spatial refinement when small details need inspection

Stop when targets are verified or the search budget is exhausted

Pass selected frames and timestamps to the answer model

T*: Evaluate the Search Itself

Measure	Question it answers
Temporal precision / recall	Do selected frames cover annotated moments within a 5 s threshold?
Visual similarity	Do selected frames resemble reference evidence?
Answer accuracy	Does the downstream model answer correctly?
Total cost	What do search and answering cost together?

A high answer score can hide poor or redundant evidence selection

T*: Results Under a 32-Frame Answer Budget

LongVideoBench XL, 15–60 minute videos, without subtitles

Answer model	Uniform sampling	With T*
GPT-4o	50.5%	53.1%
LLaVA-OneVision-72B	56.5%	62.4%

Query-dependent frame selection improves QA in this setting

Search incurs additional cost beyond the 32 frames passed to the answer model

T*: The Assumption Behind Efficient Search

The question should expose visual targets or cues that the search model can recognize

A small set of frames must retain the evidence needed for the answer

Failure cases: nearly identical events, tiny targets, or questions requiring continuous motion

Questions for Discussion

Videos are often long and redundant, making them difficult to process.
What similar challenges arise in other data modalities?

When would adding more frames make a model less reliable? How would you test your explanation?

For an hour-long robot recording, what should remain in memory?
How should you decide which details could safely be discarded?